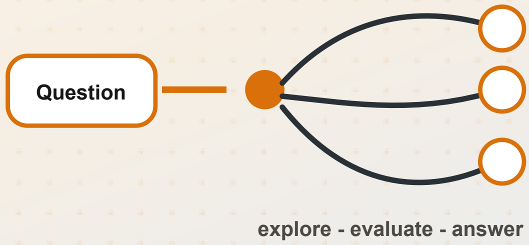
EPISODE 09

Can AI really think?

Direct generation and reasoning-oriented generation



AI NAMASTE AI - SEASON 1

Episode 09

Can AI really think?

The season finale explores what “thinking” means, why fluent language models can fail simple problems, how reasoning models use reinforcement learning and inference-time computation, and why powerful machine reasoning still does not settle the philosophical question of human-like thought.

02:46:46 | WebVTT transcript only

In this episode

- **01** Thinking, reasoning, and fluent generation
- **02** Direct answers versus deliberate computation
- **03** DeepSeek-R1, AlphaGo, and reinforcement learning
- **04** Intermediate reasoning and chain of thought
- **05** Inference-time compute and overthinking
- **06** Verifiable rewards and three kinds of evaluator
- **07** Chain, tree, and graph-of-thought structures
- **08** The limits of reasoning-and the question left open

Pause before you answer

00:00 – 02:28

Can AI really think?

Before explaining models, the instructor asks a harder question:

> What do we actually mean when we say that a person is thinking?

The learner is invited to pause the video, sit with the question for a few minutes, and form an initial answer. That pause is part of the lesson. The episode does not begin by declaring that machine reasoning is human thought. It first studies the engineering, then returns to the philosophical question at the end.

The previous episode showed that a base model is not the complete assistant. A base model is a capable form of next-token prediction. A modern assistant adds instruction following, preferences, guardrails, and tools. This finale adds another major capability: **reasoning**.

> **Q: What will count as "thinking" in the technical part of this episode?**
>
> **A:** The lecture uses thinking broadly, but focuses mainly on reasoning: intermediate decisions and computation that help the system plan, compare, check, and revise before answering.

Brilliant prose, elementary mistake

02:28 – 08:39

A strong language model can explain quantum physics. It can answer follow-up questions, write long technical prose, and produce a book-like response about DNA-related science, rocket engineering, or space technology.

Then it can fail something that looks trivial.

Demonstration 1: which decimal is bigger?

The instructor asks GPT-4 Turbo:

> What is bigger: 9.11 or 9.9?

The displayed response says **9.11 is bigger**. The instructor states that the correct comparison is **9.9 > 9.11**.

His interpretation is that the model may have fallen into the pattern "11 is greater than 9" instead of carrying out the decimal comparison.

> **WHAT WENT WRONG?**
>
> The answer was fluent and confident, but the numerical relationship was wrong. Generation quality did not guarantee calculation quality.

Demonstration 2: every third character

The next prompt is:

> Print every third character in "Namaste Artificial Intelligence."

The instructor manually checks the returned characters and concludes that the model's answer is wrong.

The exact corrected sequence is not reliable in the captions: two spoken fragments conflict with each other. **[Transcript unclear]** The supplied transcript does not preserve a trustworthy final character sequence.

Why can one model do both?

The lecture separates two abilities:

- **Generation:** continue learned language patterns fluently.
- **Reasoning:** pause, carry out intermediate work, check an answer, and make a decision.

The model has seen a vast amount of material about scientific subjects. It can generate within those patterns. A character-selection algorithm or numerical comparison may require a different kind of deliberate operation.

The instructor adds a practical qualification: a modern assistant may call a coding or calculator-like tool and solve these tasks correctly. The demonstrations isolate the limits of the language model itself, not every capability of a tool-equipped product.

Thinking is larger than reasoning

`08:39 – 13:19`

The lecture treats **thinking** as a broad umbrella:

- remembering
- processing
- planning
- comparing
- deciding

Reasoning is one critical part of that umbrella. It means using connected steps and intellect to decide what follows from the available information.

Planning a birthday trip

Imagine planning a birthday trip:

1. Do you prefer mountains or beaches?
2. Which places have you already visited?
3. What remains on your bucket list?
4. What is the budget?
5. Is the season suitable?
6. Which choice fits all of those constraints?

Remembering past travel is a kind of thinking. Selecting the destination through those connected decisions is reasoning.

Preparing a guest lecture

When invited to speak, the instructor asks:

- Is the audience made of first-year students?
- Are they graduates?
- Are they senior people in a company?
- Which topic fits them?
- How should the content change?

There is no single stored sentence that answers every invitation. The speaker uses mental bandwidth to process context and make a plan.

From this point onward, “Can AI think?” primarily becomes:

> Can AI reason?

Direct generation and reasoning-oriented generation

13:19 – 20:14

Consider the request:

> Suggest a gift for my girlfriend.

A direct answer may be:

- chocolate
- a rose
- a teddy bear
- a purse

Those are common gift patterns that appear in ordinary text.

A more thoughtful answer considers:

- interests
- profession
- favorite color
- place and age
- personal preferences
- several possible gifts
- which option feels most personal

The instructor focuses on the word **thoughtful**. A thoughtful gift carries evidence that the giver considered the person instead of selecting the first familiar item.

Direct generation

The prompt goes into the model and a likely answer comes out quickly.

Reasoning-oriented generation

The system performs intermediate work, explores options, validates them, and then selects or revises an answer.

Reasoning is not presented as a completely separate, mystical operation. The intermediate work can itself be generated:

1. identify requirements
2. propose approaches
3. compare them
4. reject weak options
5. check the remaining option
6. answer

Two generation styles

The lecture distinguishes immediate continuation from deliberate intermediate computation

Direct generation

- question arrives
- model continues directly
- answer appears quickly
- useful for simple prompts

Reasoning-oriented generation

- question arrives
- intermediate computation
- steps are evaluated
- answer follows reflection

Direct generation answers immediately; reasoning-oriented generation performs intermediate work before committing.

When the extra steps change the answer

20:14 – 23:49

The 20% up, 20% down problem

> A company grows revenue by 20% and then loses 20%. Is it back to the original revenue?

Fast intuition says that +20 and -20 cancel.

Step-by-step reasoning says:

1. Begin with 100.
2. Increase by 20% → 120.
3. Decrease 120 by 20% → 96.
4. The final value is not 100.

The second percentage acts on a different base.

But do not reason deeply about everything

If the user asks:

> Translate “hello” into Hindi.

the instructor wants a direct answer. The system should not spend 30 seconds comparing the word across Punjabi, Tamil, Malayalam, Kannada, French, and Spanish.

The balance is:

- familiar, direct task → answer quickly
- multi-step or constrained task → calculate, reflect, validate, then answer
- simple task → do not waste compute by overthinking

> **A USEFUL RULE FROM THE LECTURE**
> Reasoning is valuable when intermediate computation changes the probability of getting the answer right. It is overhead when the answer is already simple and known.

Reasoning happens during inference

23:49 – 29:13

A reasoning model attempts to bring deliberate computation into **inference**.

Training has already happened. At inference time:

1. the user gives a prompt
2. the model performs a forward computation
3. instead of immediately finalizing the answer, it can generate intermediate steps
4. it returns a final response

Debugging line 15

User:

> There is an error on line 15.

A thoughtless answer might be:

> Remove line 15.

A reasoning-oriented response breaks the problem down:

1. Read line 15.
2. Inspect nearby code.
3. Check for a syntax error.
4. Check for a logical error.
5. Read the actual error message.
6. Form possible causes.
7. Test a repair.
8. Return the corrected result.

Debugging without reasoning is almost a contradiction. The system must inspect evidence and narrow down a cause.

> **BEHIND THE SCENES**
> The lecture describes reasoning as additional generation organized into smaller, dependent subproblems. It is not presented as completely separate from generation.

The DeepSeek-R1 moment

`29:13 – 35:12`

The instructor introduces **DeepSeek-R1**, a reasoning model released by a Chinese company. He presents its release as an important moment that caused major technology-market turmoil and intensified US-China competition.

The research-paper title spoken in the lecture is:

> DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

The paper is described as showing that reasoning ability can be intensified through **pure reinforcement learning**, without requiring human-labeled reasoning trajectories in the expected way.

> **SOURCE BOUNDARY**
> The transcript calls this a historic public breakthrough and links it to a market decline. No complete publication date or cited market source is supplied, so the statements remain attributed to the lecture.

Reinforcement learning recap

The basic loop is:

1. The model tries something.
2. A good result receives a positive reward.

3. A bad result receives a penalty or negative reward.
4. The parameters are updated.
5. The model tries again.

DeepSeek-R1 did not invent reinforcement learning. The instructor immediately travels backward to an older and very visual example.

AlphaGo and self-play

35:12 – 49:40

The lecture revisits AlphaGo, developed by Google DeepMind, and its 2016 matches against Lee Sedol.

The instructor states:

- AlphaGo won the five-game match **4-1**.
- **Move 37** surprised Lee Sedol and the commentators.
- Lee Sedol paused for about 15 minutes.
- The move was outside the familiar patterns expected by the human champion.

The transcript first calls Move 37 something never played in history, then softens the statement: somebody may have played it, but it was not a famous move.

Two sources of learning

The instructor reads two important ingredients from the displayed material:

- **supervised learning from human expert moves**
- **reinforcement learning from self-play**

In self-play:

1. AI plays against AI.
2. The game produces a winner and a loser.
3. A win gives a positive signal.
4. A loss gives a negative signal.
5. The network updates.
6. The system plays again and again.

One person can experience only a finite number of games. A machine can simulate far more. It can learn patterns from its own trials instead of waiting for a human expert to label every move.

Beyond the human-demonstration ceiling

The displayed graph is described conceptually:

- supervised learning improves toward the capability present in human expert demonstrations
- reinforcement learning from self-play continues beyond that level
- improvement eventually flattens into a plateau

The lecture does not say improvement is infinite. The plateau becomes an important limit later.

> **INSTRUCTOR UNCERTAINTY**
>
> The instructor says, “I think,” when discussing whether Demis is still associated with Google. That employment claim should not be treated as established.

Research should create curiosity

`49:40 – 01:03:07`

The AlphaGo and DeepSeek discussion becomes a short lesson in how to approach research:

- read the introduction
- read the conclusion
- ask for a summary if the mathematics is too difficult
- explore more deeply when curiosity and time allow
- do not assume one course can cover every paper

The instructor’s goal is not to reproduce the entire AlphaGo or DeepSeek paper. It is to make the learner curious enough to investigate.

The geopolitical tangent

The lecture says:

- the DeepSeek release wiped more than one trillion dollars from global technology stocks
- US and Chinese companies are competing intensely
- DeepSeek-R1 is open and downloadable
- some US companies may access Chinese-origin models through a US provider such as Together.ai

These are broad, partly speculative statements in the VTT. The instructor himself says he does not fully know the political or data-routing situation.

The captions render the DeepSeek chat address as:

> **INCOMPLETE OR UNCLEAR URL**
>
> `chat.deepsea.com` is what the transcript contains. It may be a speech-to-text error, and the exact URL remains unresolved. **[Transcript unclear]**

The other spoken site is `together.ai`.

More steps, more accuracy-and more tokens

A displayed graph is described as relating:

- number of reasoning steps
- answer accuracy
- average response length

The instructor says accuracy rises as the model takes more reasoning steps and that the curve crosses a dotted human-participant line. He also says response length and token consumption rise.

The task, axes, and exact benchmark are not fully contained in the transcript.

A travel plan that thinks before it answers

01:03:07 – 01:11:09

The instructor opens DeepSeek, selects an expert model with deep thinking, and gives it a constrained task:

> I want to go on a vacation in Malaysia in October, with a budget of ₹2 lakh, with my wife and an infant. I want to explore cool things to eat. Give me an itinerary and a price breakdown.

This prompt has no single stored answer. It contains dependencies:

- destination
- month
- budget
- two adults
- an infant
- food
- itinerary
- pricing

What the visible reasoning considers

The interface appears to ask itself about:

- which language to answer in
- a reasonable trip duration, because none was specified
- candidate cities such as Kuala Lumpur and Penang, plus another place whose name is unclear in the captions
- seasonal rain
- visa requirements
- currency conversion
- flight, hotel, food, and local-transport budgets
- multiple itinerary options
- infant-friendly details
- baby food, spice level, and bottled water

It reportedly thinks for **116 seconds** before producing a separate, polished answer.

> **MISSING VISIBLE DETAILS**
> The full itinerary, prices, current visa information, and tool calls are not captured in the VTT. **[VIDEO FRAME NEEDED]** Suggested timestamp: 01:05:16.

The instructor compares three levels:

- no deep thinking: fast response, narrated as a 12-day/11-night plan with more assumptions
- instant thinking: about 20 seconds
- expert deep thinking: about 116 seconds

The demonstration supports one claim:

> For a constrained research task, spending useful compute during inference can produce a more considered answer.

It does not establish that the resulting travel advice is correct. The instructor himself says he has not verified how good the plan is.

Research papers behind assistant behavior

01:11:09 – 01:20:18

Two earlier papers are named:

1. **Fine-Tuning Language Models from Human Preferences** - described as a 2020 OpenAI paper.
2. **Training Language Models to Follow Instructions with Human Feedback** - described as a 2022 OpenAI paper.

They connect this finale to the previous episode:

- supervised fine-tuning
- human preferences
- instruction following
- reinforcement learning
- the transition from base model to assistant

The instructor then performs a live query about the authors' current employment, including Dario and Anthropic. The exact table, sources, author list, and date are missing from the VTT. Some records are explicitly described as ambiguous.

> **DO NOT CONVERT THE LIVE QUERY INTO COURSE FACT**
> The narration says none of the listed authors remains at OpenAI and several joined Anthropic, but the visible evidence is absent.

The durable lesson from this tangent is editorial rather than biographical: a small group of researchers and a few important papers can reshape the behavior of widely used assistants.

Intermediate reasoning

01:20:18 – 01:25:33

The lecture returns to a clean arithmetic example:

> There are 23 students. Each needs 4 notebooks. Notebooks are sold only in packets of 10. How many packets should the teacher buy?

1. 23 students × 4 notebooks = 92 notebooks.
2. 92 ÷ 10 = 9.2 packets.
3. A partial packet cannot be purchased.
4. Round up to 10 complete packets.

The answer is **10 packets**.

Each step preserves information needed by the next. This is **intermediate reasoning**.

The important timing distinction is that the model is already trained. These steps are generated on the fly while answering the user-during inference.

Chain of thought

01:25:33 – 01:30:07

The paper title spoken is:

> Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

The lecture attributes it to Google research/Brain/DeepMind and describes it as a 2023 paper. Exact bibliographic details are not supplied.

A **chain of thought** is defined as a sequence of intermediate reasoning steps before the final answer.

The system:

1. breaks the original problem into smaller questions
2. solves one subproblem
3. passes that result to the next subproblem
4. continues the chain
5. combines the work into an answer

The travel-planning example forms a chain across budget, visa, locations, infant needs, transport, and food.

The instructor says the discussed research shows that intermediate reasoning improves results on complex tasks. He later qualifies the idea: more thinking is not infinitely useful.

A bat, a ball, and the cost of answering too quickly

01:30:07 – 01:34:23

> A bat and a ball together cost 110. The bat costs 100 more than the ball. How much does the ball cost?

The immediate answer many people give is 10.

Let the ball cost **x**.

```
ball = x
bat = 100 + x

x + (100 + x) = 110
2x + 100 = 110
2x = 10
x = 5
```

The ball costs **5** and the bat costs **105**.

The problem is not difficult, but the sentence invites a quick subtraction: 110 - 100. Structured computation prevents premature commitment.

> **WHY THIS MATTERS**
>
> Humans and models can both be misled by the first fluent-looking answer. Sometimes a small amount of computation before commitment is enough to correct it.

A second scaling dimension: inference-time compute

01:34:23 – 01:44:42

Earlier improvements concentrated on training:

- better pre-training data
- more model parameters
- more training compute
- supervised fine-tuning
- human feedback
- reinforcement learning

Reasoning models open another dimension:

> Spend more useful compute solving this particular prompt during inference.

For a difficult question, a system may:

- generate intermediate steps
- explore alternatives
- verify calculations
- call tools
- write and execute code
- evaluate candidate answers
- revise a draft
- check constraints
- compile a final response

Training-time compute

Builds the model and its broad capabilities before the user asks a question.

Inference-time compute

Works on the current prompt after training is complete.

The instructor summarizes:

> A model's intelligence is not determined only by how much computation was used during training. Useful computation performed while answering can matter too.

Two places to spend computation

Reasoning models open an additional capability dimension during inference

Training time

- data shapes parameters
- loss guides updates
- capability is learned
- a model is produced

Inference time

- a prompt is received
- reasoning steps are generated
- more compute may be spent
- a final answer is produced

Training-time compute shapes the model; inference-time compute is spent reasoning about the current prompt.

More tokens are not automatically more intelligent

01:44:42 – 01:52:13

Ask:

> What is 5 + 5?

The answer is 10. Spending two minutes on it does not create a better answer.

The instructor imagines forced overthinking sending the model into an irrelevant JavaScript interpretation and producing “55.” No types or string-concatenation context is supplied, so this is only an illustration-not a complete JavaScript claim.

The three-zone mental model

1. **Underthinking:** too little compute can produce a premature answer.
2. **Useful thinking:** enough computation can improve difficult reasoning.
3. **Overthinking:** more steps bring diminishing returns and may even abandon an initially correct solution.

More reasoning can increase the probability of success. It does not convert probability into certainty.

The system must learn how much reasoning a problem deserves.

Teaching reasoning through rewards

01:52:13 – 01:58:49

The reinforcement-learning loop returns:

model behavior → **evaluator** → **positive reward or penalty** → **parameter update** → **new behavior**

This works especially well when the task is **verifiable**.

A programming task

The instructor asks for a JavaScript program related to a palindrome number. The model can produce many candidates. Each candidate can be run against tests:

- correct output → reward
- incorrect output → penalty

Over many attempts, patterns associated with passing tests become stronger.

> **TASK DEFINITION IS UNCLEAR**
>
> The prompt says "palindrome number," but examples such as `123 -> 321` and `432 -> 234` describe digit reversal. The exact programming task is internally inconsistent. **[Transcript unclear]**

The idea extends to:

- heap sort
- data-structure and algorithm problems
- traveling-salesman-like problems
- mathematical questions with known answers
- any code that can be executed against tests

A subjective task

Now ask the model to:

- write the best poem
- produce the funniest joke
- explain something in the most helpful way
- plan the best Malaysia itinerary
- give a medical response that requires expertise

There is no single deterministic expected string. A human or learned judge must evaluate quality.

RLVR: verifiable rewards

`01:58:49 – 02:02:40`

The episode names **RLVR**:

> Reinforcement learning with verifiable rewards.

Preference-based feedback

Useful for subjective tasks. Humans or reward models compare tone, style, usefulness, or taste.

Verifiable rewards

Useful when a rule, test, calculation, or game outcome can determine correctness automatically.

AlphaGo fits the second side:

- a completed game has a winner
- the result is automatically available
- self-play can continue without a human judging every move

Mathematics and programming are highlighted because answers or behavior can often be checked.

Two reward sources in the episode

Both use reinforcement learning, but their feedback is obtained differently

RLHF	RLVR
humans compare responses	the answer can be checked
preferences become feedback	reward uses verifiable result
a reward model may approximate	math and code examples
subjective evaluation remains	feedback is less subjective

RLHF derives feedback from human preferences or their learned approximation; RLVR uses an answer that can be checked.

Who judges the output?

02:02:40 – 02:05:51

An **evaluator** is the judge that decides whether output is good or bad.

The lecture names three kinds.

1. Deterministic evaluator

Use an exact rule:

- compare 5 + 5 with 10
- run code against tests
- check who won a game

The instructor says to prefer this whenever the task permits it.

2. Human evaluator

A person reads or tests the answer.

Useful when:

- taste matters
- interpretation matters
- expertise is required
- no exact answer exists

Examples include poetry, jokes, essays, and medical review.

3. Model evaluator

Another language model judges the result.

This is commonly described in the lecture as:

> LLM as a judge.

It scales more easily than asking humans to review every candidate.

LLM as a judge

02:05:51 – 02:11:12

The instructor says a model can judge dimensions such as:

- relevance
- clarity
- instruction following
- style
- completeness
- code explanation
- some factual qualities

But the judge is still a model. It can have:

- social or data bias
- position bias
- style preference
- a preference for long answers
- limited ability on subjective beliefs

> **VERY IMPORTANT**
> Using AI to evaluate AI scales evaluation. It does not magically create objective truth.

The evaluator must itself be evaluated:

model creates → model judges → who judges the judge?

The instructor argues that improvement eventually plateaus. In his opinion, an infinite recursion of AI training and judging AI does not automatically create movie-like AGI or superintelligence. He also admits that definitions change and future developments remain uncertain.

This is a personal position in the lecture, not a settled factual conclusion.

Chain, tree, and graph of thoughts

02:11:12 – 02:18:48

Chain of thought

One path:

A → B → C → answer

Each step depends on the previous step. If step B is wrong, the error can spread through the rest of the chain.

The instructor jokingly compares it with a linked list.

Tree of thought

A tree explores alternatives:

- branch into several approaches
- continue promising branches
- backtrack when one fails
- compare successful candidates
- select the strongest answer

This helps when one line of reasoning becomes stuck or wrong.

Graph of thoughts

A graph allows branches to reconnect and combine useful pieces.

Heap-sort illustration:

- Approach A finds an efficient **$O(n)$** method.
- Approach B catches an important edge case but uses **$O(n^2)$** work.
- A later step combines the efficiency from A with the edge-case handling from B.

The complexities are illustrative; no executable heap-sort code is supplied.

Chain

Explore one dependent path.

Tree

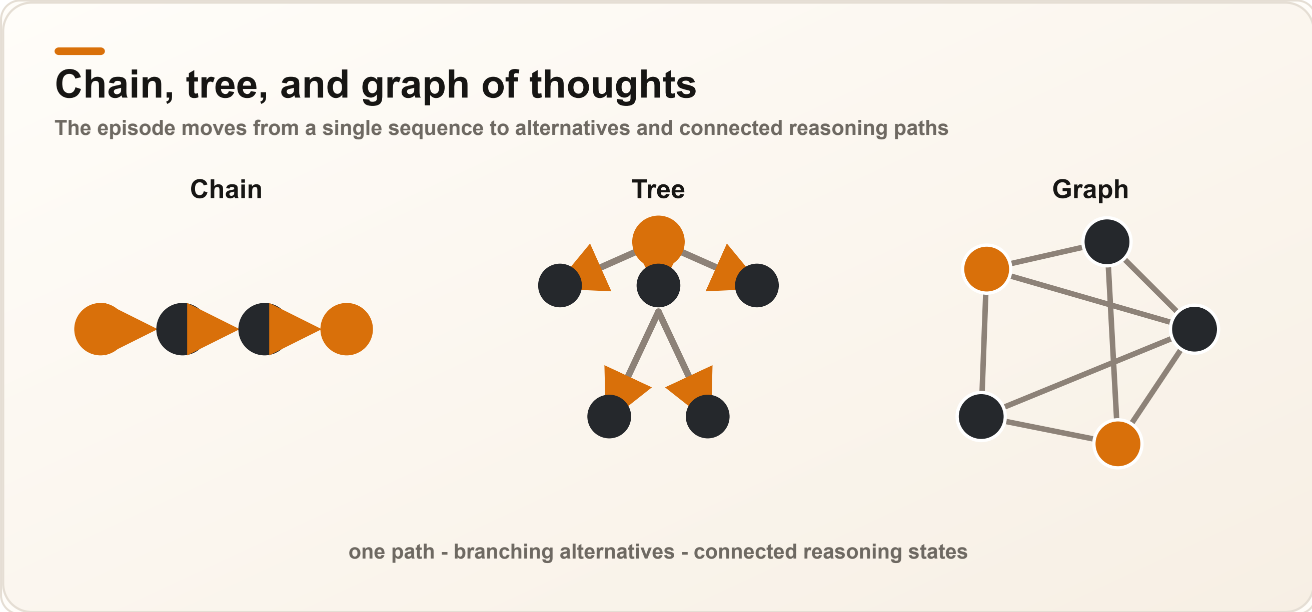
Explore alternatives and backtrack.

Graph

Explore alternatives, reconnect branches, and combine useful ideas.

Final response

Select one candidate or merge the strongest parts into a polished answer.



A chain follows one path, a tree explores alternatives, and a graph can reconnect and combine useful reasoning states.

Visible reasoning is not necessarily faithful

`02:18:48 – 02:26:49`

DeepSeek displays an English “thought process.” Is that prose a literal transcript of the model’s internal computation?

The instructor says: not necessarily.

- Internally, the model is performing mathematical computation.
- The visible English is generated around that computation.
- The model may work on several subproblems in parallel even if the displayed trace appears sequential.
- The explanation may be simplified, summarized, reconstructed, or partly post hoc.
- Branches may be merged after separate work.

The lecture refers to Anthropic research showing cases where generated chain-of-thought text did not fully match the information influencing the answer. No exact paper title or link is supplied.

A human analogy

Why did you choose one particular word while speaking?

Sometimes you can explain. Sometimes the word appeared through a learned association and you cannot reconstruct the path. The instructor uses his own spontaneous “linked list of thought” joke as an example.

Visible explanation and causal process are not always identical-for humans or models.

Why some providers hide raw traces

The instructor contrasts:

- DeepSeek’s long visible trace

- shorter “thinking” indicators or summaries in other assistants

He paraphrases reasons given by OpenAI:

- preserve usefulness for monitoring
- avoid exposing unaligned intermediate content
- protect user experience
- account for competitive considerations

He then expresses skepticism and speculates about politics, competition, and distillation. Those parts are explicitly his opinion.

Reasoning models are still fallible

02:26:49 – 02:28:48

Reasoning models can still:

- hallucinate
- make arithmetic errors
- misunderstand the question
- accept a false assumption
- misuse tools
- overthink
- fail on an unfamiliar problem

More thinking can improve the chance of success. It cannot guarantee correctness.

The instructor points to interface language such as “AI generated; use for reference only.” He warns against making medical decisions solely from an assistant answer. More context and deeper reasoning may change the answer, but neither provides a guarantee.

Match the capability to the task

02:28:48 – 02:35:39

The final technical recap maps tasks to capabilities.

Task	Capability emphasized in the lecture
Explain recursion	Language generation from learned knowledge
Complex arithmetic	Calculator or coding tool
Find the current stock price	Current-data or web-search tool
Solve a mathematical proof	Reasoning
Count characters	Code/tool use
Summarize an essay	Generation
Search a private document	Retrieval plus generation
Debug a program	Reasoning plus coding tools plus generation

The lecture briefly introduces **retrieval-augmented generation (RAG)** as a future-season topic: retrieve information, then use it to augment generation.

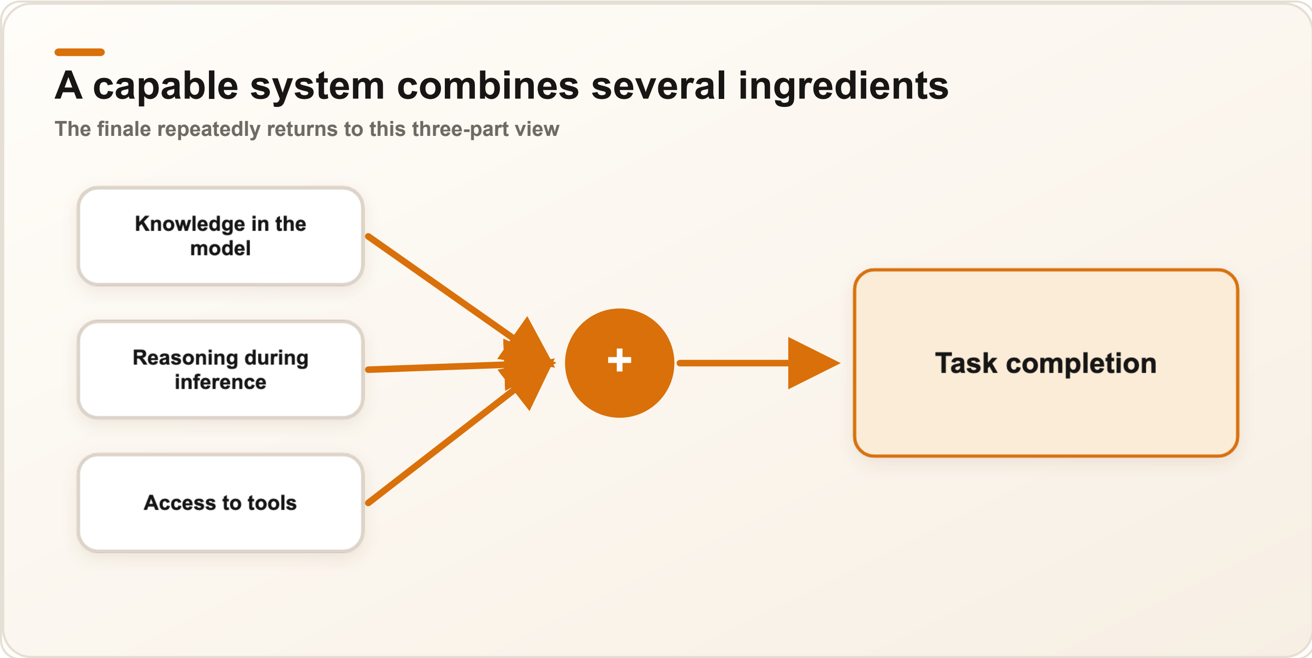
The modern assistant has three broad powers

1. **Knowledge** from training.
2. **Reasoning** from post-training, reinforcement learning, and inference-time computation.
3. **Tools** for current data, calculation, code, retrieval, images, video, and other actions.

The Malaysia task uses all three:

- knowledge suggests destinations
- reasoning adapts the plan to budget and infant needs
- tools may check current visa information, currency, or flights

The full product is much more than the literal phrase “Generative Pre-trained Transformer.”



The finale’s three-part view: learned knowledge, reasoning during inference, and external tools work together to complete a task.

So, can AI really think?

02:35:39 – 02:46:45

From an engineering perspective, machine behavior has become remarkable. A reasoning-capable assistant can:

- decompose a problem
- compare alternatives
- plan
- validate constraints
- revise a solution
- use tools
- research
- debug

That behavior can look like thinking.

The instructor then changes the lens.

Machine reasoning remains mathematical computation. Human thought is also shaped by:

- emotions
- personal upbringing
- memory
- culture
- location
- bias

- lived experience
- changing mood

Humans are inconsistent. What a person thinks today may change tomorrow-or five minutes later.

Picture a pet animal

Ask several people to imagine a pet.

One person may picture:

- a cow
- a buffalo
- a horse
- an elephant
- a cat
- a dog

The association depends on place and life experience. Human thinking is personal and subjective.

> **Q: Does powerful reasoning settle the meaning of thought?**
>
> **A:** Not for the instructor. Humans have capabilities machines do not; machines have capabilities humans do not. Their strengths are different, and the lecture leaves the final answer to the learner.

The instructor refuses to force the conclusion into one word.

> Can AI really think?

There is no final “yes” or “no” in this episode. The technical mechanisms are explained. The philosophical answer remains open.

Chapter summary

The episode began with a paradox: a model can produce beautiful scientific prose and still fail a decimal comparison or character-selection task. Fluent generation is not the same as deliberate checking.

Thinking is broad; reasoning is the part of the lesson concerned with intermediate decisions and computation. Direct generation is useful for simple familiar tasks. Reasoning-oriented generation helps with planning, debugging, research, calculations, and other prompts whose constraints must be preserved.

DeepSeek-R1 and AlphaGo illustrate reinforcement learning in two settings. AlphaGo’s self-play uses automatically verifiable win/loss outcomes. Modern reasoning models spend additional compute during inference, breaking one prompt into subproblems, exploring alternatives, checking constraints, and revising an answer.

Intermediate reasoning can take the form of a chain, tree, or graph. Yet more reasoning is not always better: overthinking can create diminishing returns. Verifiable tasks can use deterministic rewards; subjective tasks need humans or model judges. LLM-as-a-judge scales evaluation but inherits bias and does not create objective truth.

Visible English reasoning may not be a faithful transcript of internal computation, and reasoning models still make mistakes. A modern assistant combines knowledge, reasoning, and tools-but whether this counts as human-like thought remains a question for the learner.

Key takeaways

- Fluent generation does not guarantee correct calculation or deliberate checking.
- The episode uses reasoning as the technical core of the broader word "thinking."
- Direct generation is efficient for simple tasks; deliberate reasoning helps when dependencies and constraints matter.
- Reasoning models spend useful compute during inference, after training is complete.
- Reinforcement learning improves behavior through attempts, rewards or penalties, and parameter updates.
- AlphaGo self-play demonstrates learning from a verifiable game outcome at scale.
- Intermediate reasoning breaks a problem into steps before the final answer.
- More reasoning can raise the probability of success, but it does not guarantee correctness.
- Underthinking, useful thinking, and overthinking require different compute budgets.
- RLVR uses automatically verifiable rewards for tasks such as mathematics, code, and games.
- Evaluation can be deterministic, human, or model-based.
- LLM-as-a-judge scales review but carries bias and still needs evaluation.
- Chains follow one path, trees branch and backtrack, and graphs can recombine useful ideas.
- A visible reasoning trace may be a reconstruction rather than a literal internal record.
- Modern assistants combine knowledge, reasoning, and tools.
- The lecture deliberately leaves "Can AI really think?" unanswered.

Revision questions

1. Why does the instructor ask the learner to define thinking before discussing AI?
2. What happened in the 9.11-versus-9.9 demonstration?
3. Why is the final result of the every-third-character example uncertain in the supplied transcript?
4. How does the lecture distinguish generation from reasoning?
5. Which activities sit under the broad term thinking?
6. How do the birthday-trip and guest-lecture examples illustrate reasoning?
7. What is the difference between a direct gift suggestion and a thoughtful one?
8. Why does +20% followed by -20% produce 96 when the starting value is 100?
9. Which kinds of prompt should receive a direct answer?

10. What does inference mean in this chapter?
11. How does the line-15 debugging example decompose a problem?
12. What role does reinforcement learning play in the DeepSeek-R1 discussion?
13. Which two learning sources are named for AlphaGo?
14. Why can self-play scale beyond human demonstration data?
15. What does a win or loss provide during AlphaGo training?
16. What constraints are present in the Malaysia travel prompt?
17. What kinds of subproblem does the visible travel reasoning explore?
18. What is intermediate reasoning?
19. How does the notebook-packet problem arrive at 10 packets?
20. What is a chain of thought?
21. Why is 5-not 10-the answer to the bat-and-ball problem?
22. What is the difference between training-time and inference-time compute?
23. Why do more tokens not automatically mean more intelligence?
24. What are underthinking, useful thinking, and overthinking?
25. What makes a task verifiable?
26. What does RLVR stand for?
27. Why are mathematics, code, and games useful for verifiable rewards?
28. What are the three evaluator types?
29. What is LLM as a judge?
30. Why must a model evaluator itself be evaluated?
31. How do a chain, tree, and graph of thoughts differ?
32. Why might a visible English thought process not be faithful to internal computation?
33. What limitations remain in reasoning models?
34. Which task examples in the recap use generation, tools, retrieval, reasoning, or a combination?
35. What three capabilities make up the final modern-assistant picture?
36. Why does the instructor leave the central question without a yes-or-no answer?

One last thing

This is the finale of the season. The instructor closes by inviting the learner to keep the question alive: after understanding the mechanics, decide for yourself whether what the machine does should be called thinking.